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3 Complex Analysis 2024
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Test 1
 28 August 2024



Instructions:

- This question paper consists of **5 pages** (including this one).
- You have 90 minutes to answer the questions.
- Answer all questions in the spaces provided on this question paper. Use the backs of pages if needed.
- Use your *UCT Examination Answer Book* for rough work. The work in your UCT Examination Answer Book will not be marked. Only the answers that you write on this question paper will be marked.
- Give full proofs.
- Be careful to provide answers that we can read and make sense of at all times. If we can't read your handwriting, we will mark your solution incorrect.
- 40 points = 100% (42 points available).

Problem	1	2	3	4	Σ	Mark
Maximum Points	12	10	10	10	40	100%
Result						

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Problem 1

(12 points)

- Let $G \subseteq \mathbb{C}$ be an open set and $f : G \rightarrow \mathbb{C}$. Let $z \in \mathbb{C}$. Define what is meant by f being differentiable at z , and define the derivative of f at z .
- Show by definition that the function $f(z) = \bar{z}^2$ is differentiable at $z = 0$, and determine $f'(0)$.
- Show by a method of your choice that $f(z) = \bar{z}^2$ is not differentiable at any $z \neq 0$.

Solution.

- f is differentiable at z if the limit $\lim_{h \rightarrow 0} \frac{f(z+h) - f(z)}{h}$ exists. In this case, the derivative of f at z is defined by

$$f'(z) = \lim_{h \rightarrow 0} \frac{f(z+h) - f(z)}{h}.$$

- We have

$$\lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0} \frac{\bar{h}^2}{h}.$$

We claim that $\lim_{h \rightarrow 0} \frac{\bar{h}^2}{h} = 0$. Indeed, using $|\bar{h}| = |h|$ we obtain

$$\lim_{h \rightarrow 0} \left| \frac{\bar{h}^2}{h} - 0 \right| = \lim_{h \rightarrow 0} \frac{|h|^2}{|h|} = \lim_{h \rightarrow 0} |h| = 0.$$

Thus, f is differentiable at $z = 0$, and $f'(0) = 0$.

- We are going to use the Cauchy-Riemann equations $u_x = v_y$, $u_y = -v_x$ that give a necessary condition for differentiability. We have

$$f(x+iy) = (x-iy)^2 = x^2 - 2ixy - y^2,$$

so that $u(x, y) = x^2 - y^2$ and $v(x, y) = -2xy$. The derivatives are

$$u_x = 2x, \quad u_y = -2y, \quad v_x = -2y, \quad v_y = -2x.$$

The Cauchy-Riemann equations take the form

$$2x = -2x, \quad -2y = 2y.$$

The only solution is $(x, y) = (0, 0)$. It follows that f is not differentiable at any $z \neq 0$.

Problem 2

(10 points)

Consider the power series $\sum_{n=1}^{\infty} \frac{z^{2n}}{n^2}$.

- (a) Determine the radius of convergence R of this series.
 (b) Show that the series converges uniformly in the disc $\{z \in \mathbb{C} : |z| \leq R\}$.

Solution.

- (a) *1st solution.* We are going to use the Ratio Test for the real series $\sum_{n=1}^{\infty} \frac{|z|^{2n}}{n^2}$. We have

$$L = \lim_{n \rightarrow \infty} \frac{|z|^{2n+2}}{(n+1)^2} \frac{n^2}{|z|^{2n}} = |z|^2 \lim_{n \rightarrow \infty} \left(\frac{n}{n+1} \right)^2 = |z|^2 < 1 \iff |z| < 1.$$

The radius of convergence of the series $\sum_{n=1}^{\infty} \frac{z^{2n}}{n^2}$ is $R = 1$.

2nd solution. The coefficients are

$$a_m = \begin{cases} \frac{1}{\left(\frac{m}{2}\right)^2} = \frac{4}{m^2}, & m \text{ even,} \\ 0, & m \text{ odd.} \end{cases}$$

For the radius of convergence we have

$$\frac{1}{R} = \limsup_{m \rightarrow \infty} |a_m|^{\frac{1}{m}} = \lim_{m \rightarrow \infty} \left(\frac{4}{m^2} \right)^{\frac{1}{m}} = \lim_{m \rightarrow \infty} e^{\frac{1}{m} \ln \left(\frac{4}{m^2} \right)} = \lim_{m \rightarrow \infty} e^{\frac{1}{m} (\ln 4 - 2 \ln m)} = e^0 = 1.$$

Thus, $R = 1$.

- (b) Let us show that the series converges uniformly in the disc $\{z \in \mathbb{C} : |z| \leq 1\}$. We apply the Weierstrass M-Test. For $|z| \leq 1$ we also have $|z|^{2n} \leq 1$, and thus

$$\left| \frac{z^{2n}}{n^2} \right| \leq \frac{1}{n^2} \quad \text{for all } n \in \mathbb{N}, |z| \leq 1.$$

We know that the series $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges. By the Weierstrass M-Test, the series $\sum_{n=1}^{\infty} \frac{z^{2n}}{n^2}$ converges uniformly (and absolutely) in the disc $\{z \in \mathbb{C} : |z| \leq 1\}$.

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Problem 3

(10 points)

Recall that $\cos z = \frac{e^{iz} + e^{-iz}}{2}$, $\sin z = \frac{e^{iz} - e^{-iz}}{2i}$, $z \in \mathbb{C}$.

- (a) Prove that $\cos(2z) = \cos^2 z - \sin^2 z$.
- (b) Find one solution of the equation $\cos z = i$.

Solution.

- (a) We have

$$\begin{aligned}
 \cos^2 z - \sin^2 z &= \left(\frac{e^{iz} + e^{-iz}}{2} \right)^2 - \left(\frac{e^{iz} - e^{-iz}}{2i} \right)^2 \\
 &= \frac{1}{4} (e^{i2z} + 2e^{iz}e^{-iz} + e^{-i2z}) + \frac{1}{4} (e^{i2z} - 2e^{iz}e^{-iz} + e^{-i2z}) \\
 &= \frac{1}{4} (2e^{i2z} + 2e^{-i2z}) = \frac{e^{i2z} + e^{-i2z}}{2} = \cos(2z).
 \end{aligned}$$

- (b) Let $z = x + iy$, then

$$\begin{aligned}
 \cos z &= \frac{1}{2} (e^{iz} + e^{-iz}) = \frac{1}{2} (e^{i(x+iy)} + e^{-i(x+iy)}) = \frac{1}{2} (e^{ix}e^{-y} + e^{-ix}e^y) \\
 &= \frac{1}{2} (e^{-y}(\cos x + i \sin x) + e^y(\cos x - i \sin x)) \\
 &= \frac{1}{2} (\cos x (e^{-y} + e^y) + i \sin x (e^{-y} - e^y)).
 \end{aligned}$$

The equation $\cos z = i$ is equivalent to

$$\frac{1}{2} \cos x (e^{-y} + e^y) = 0, \quad \frac{1}{2} \sin x (e^{-y} - e^y) = 1.$$

We take, for example, $x = \frac{\pi}{2}$. Then the first equation is fulfilled, and the second equation takes the form

$$e^{-y} - e^y = 2.$$

We denote $t = e^y$. Notice that $t > 0$ and $e^{-y} = \frac{1}{t}$. The above equation is

$$\frac{1}{t} - t = 2.$$

This is the same as

$$t^2 + 2t - 1 = 0.$$

The solutions of the above quadratic equation are $-1 - \sqrt{2}$ and $-1 + \sqrt{2}$. Since in our case $t > 0$, we take $t = e^y = -1 + \sqrt{2}$. Thus, $y = \ln(\sqrt{2} - 1)$.

Summarizing, a solution of the equation $\cos z = i$ is $z = \frac{\pi}{2} + i \ln(\sqrt{2} - 1)$.

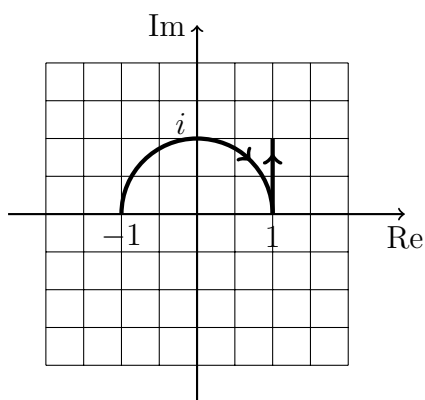
Problem 4

(10 points)

Consider the path

$$\gamma(t) = \begin{cases} e^{-i\pi t}, & t \in [-1, 0], \\ 1 + it, & t \in [0, 1]. \end{cases}$$

- (a) Sketch the path in the graph paper below. Don't forget to label the axes and to show the orientation of the path.
- (b) Calculate $\int_{\gamma} z \, dz$.
- (c) Calculate $\int_{\gamma} \bar{z} \, dz$.

**Solution.**

- (a) See the graph above.
- (b) For the function $F(z) = \frac{1}{2}z^2$ we have $F'(z) = z$. Thus, by the Fundamental Theorem of Calculus,

$$\begin{aligned} \int_{\gamma} z \, dz &= F(\gamma(1)) - F(\gamma(-1)) = \frac{1}{2} \left((1+i)^2 - (-1)^2 \right) \\ &= \frac{1}{2}(1 + 2i - 1 - 1) = -\frac{1}{2} + i. \end{aligned}$$

- (c) We have

$$\begin{aligned} \int_{\gamma} \bar{z} \, dz &= \int_{-1}^0 \overline{e^{-i\pi t}} (-i\pi) e^{-i\pi t} \, dt + \int_0^1 \overline{1+it} i \, dt \\ &= -i\pi \int_{-1}^0 e^{i\pi t} e^{-i\pi t} \, dt + i \int_0^1 (1-it) \, dt \\ &= -i\pi \int_{-1}^0 1 \, dt + i \int_0^1 1 \, dt + \int_0^1 t \, dt \\ &= -i\pi + i + \frac{1}{2} = \frac{1}{2} + i(1-\pi). \end{aligned}$$